MIL-B-24911
Amendment-2
16 July 1987

SUPERSEDING
Amendment-1
4 June 1984

MILITARY SPECIFICATION

BOOTS, SAFETY, AIRCREW

This amendment forms a part of Military Specification MIL-B-24911, dated 1 March 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete title and substitute "BOOTS, FLYERS".

1.1, line 2 - Delete "aircrew" and substitute "flyers".

* line 3 - Delete "speed lacing,".

*1.2 - For widths XN, N, R, W, XW delete size "4" and substitute respectively "3, 2, 2, 2, 2". Delete all reference to size "14-1/2" and substitute "14".

PAGE 2

*2.1.1, Under SPECIFICATION, FEDERAL, - After "KK-I-570" add "NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)".

Under SPECIFICATION, MILITARY - After "MIL-C-13924" add "MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry".

Under STANDARDS, MILITARY - After "MIL-STD-129" add "MIL-STD-147 - Palletized Unit Loads".

PAGE 3

*2.2, lines 25 and 26 - After ANSI delete remaining words and substitute "American National Standard for Personnel Protection - Protective Footwear Z41-1983".

AMSC N/A FSC 8430

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PAGE 4

*3.2, line 2 - Add the following "The item will be tested and it should be a first article sample. The contracting officer shall include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.".

*3.3.1, line 13 - Delete "vamps" and substitute "uppers".

line 15 - After "cracked," add "pronounced veins,".

3.3.1.1.1 - Delete in its entirety and substitute "3.3.1.1.1 Water penetration. A minimum of 70% of the specimens tested shall withstand a minimum of 4,000 Dow Corning flexes or 15,000 Maeser flexes. A maximum of 30 percent of the specimens tested shall withstand a minimum of 1,000 Dow Corning flexes or 3,000 Maeser flexes when tested as specified in 4.3.1.".

*3.3.1.2 - Delete in its entirety and substitute "3.3.1.2 Fungicide. All leather components shall contain not less than 0.18 percent or more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.1.".

*3.3.2, lines 1, 2, 6, 7, 8 - Delete all references to "speed lacer stays" and "top band cuffs" and substitute respectively "eyelet stays" and "top band".

line 5 - Delete "paranitrophenol,".

line 7 - After "stays." add "Top band stripping of 1 inch width may be used.".

PAGE 5

*3.3.3, sentence 3 - Delete the period and add "however, any obvious variation in shade shall be matched within shoes and grouped in pairs.".

*3.3.5.1, line 3 - After "irons" add a period and delete remainder of sentence.

PAGE 6

*3.3.7.1, line 2 - Delete "4 thru 14-1/2" and substitute "2 through 14".

*3.3.9.1, after sentence 5, add "As an alternate, an equivalent fabric with a thread count of 42x34 may be used."

*3.3.9.2, line 6 - After "to" insert "cover the entire steel toe to weltline and shall".

PAGE 7

*3.3.9.5 - Delete in its entirety.

Add new paragraph after 3.3.10.1.1 - "*3.3.10.1.2 Compression resistance. A certificate of compliance shall be submitted by the manufacturer for conformance to the ANSI Z41 PT compression requirement C75.".

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PAGE 8

*3.3.10.3 and 3.3.10.4 - Delete in their entirety.

*3.3.10.8, line 4 - Delete "7" and substitute "8".

PAGE 9

*3.3.10.8.2, line 8 - Delete "1" and substitute "0".

3.3.10.9, line 7 - Delete "as specified in 3.3.10.8.2." and substitute "1/2 (± 1/16) inch longer than as specified for the steel shank in 3.3.10.8.1.".

*Table III - Delete footnote "1/" next to shoe widths and sizes. Add new column of shank size "0",

Shoe widths - XN - N "3, 3-1/2"
R - W "2, 2-1/2, 3, 3-1/2"
XW "2, 2-1/2, 3, 3-1/2"

Delete "14-1/2" wherever it appears.
After shank size 7 add "1/".
Delete footnote 1/ at end of table and substitute "1/ Shank size 7 provided for use on shoe sizes larger than 14."

*3.3.11.1, sentence 4 - Delete in its entirety and substitute "The uppers shall be sewn with black thread and the quarter linings shall be sewn with natural color thread or a shade approximating the leather lining.".

PAGE 10

*3.3.13, line 2 - Delete "72" and substitute "60".

*3.3.14.1.1, line 1 - Delete "butryal" and substitute "butyral".

line 2 - Delete "pthalate" and substitute "phthalate".

PAGE 11

*3.3.14.2 - Delete in its entirety and substitute "3.3.14.2 General adhesive. The adhesive for bonding the outsole to the welting shall be 50 psi (minimum) before and after immersion of the bonded specimens in Type III fluids of TT-S-735 for 24 (± 1/4) hours at 74.4 (± 3.6) degrees F. After immersion, the specimen shall be dried at 74.5 (± 2) degrees F and 50% (± 5%) RH for 24 (± 1/2) hours as specified in 4.3.1.".

*3.3.16, line 3 - Delete "80" and substitute "8".

*3.3.19 - After sentence 3 add "Prior to lasting, a heel reinforcement tuck may be attached to the outside heel area of the insole.".

PAGE 12

3.4,line 1 - Delete "nine" and substitute "eight".

* line 2 - Delete "with a fold down top cuff, speed lace closure with clinch clamp, "and substitute "with a full lace eyelet closure,".

* line 5 - Delete "(see figures 1 thru 4) "and substitute "(see Figure 1A).".

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PAGE 12 (Conc'd)

*3.5, Table IV - Delete "Eyelet/speed lace stay - Whole sizes, all widths
Top facing cuff - Whole sizes, two widths 2/" and substitute:
"Eyelet stay - Three sizes, one width
Top Band - No patterns provided 1/".

Delete footnotes 1/ and 2/ in their entirety and substitute: "1/ The boots shall be made with the patterns specified for MIL-B-21408, Boot, Safety. The contractor shall grade patterns smaller than size 4 in the widths required, based on the Government furnished size 4 patterns. 2/ A one inch wide stripping may be used for the top band.".

*3.6, line 5 - Add the following sentence "Upper patterns and insole patterns for sizes smaller than 4 shall be graded down by the contractor as required.".

*3.7.1, line 3 - Delete "speed lace" and substitute "eyelet".

Delete "facing cuff" and substitute "band".

PAGES 13 and 14

*3.7.3, line 1 - Between the words "The" and "cotton" insert "napped side of".

*3.7.4.2, sentence 3 - Delete in its entirety and substitute: "Unless otherwise specified, the upper stitching shall be sewn with stitch type 301 of FED-STD-751 having 8-10 stitches per inch."

Sentence 9 - After "502" insert "or 404".

Sentences 11, 13, and 15 - After "apart" add a period And delete remainder of sentences.

Sentences 17 and 18 - Delete in their entirety and substitute "The top band shall be sewn with one row of stitching before turning and two rows of stitching after turning. The two outside rows of 1/4 inch gauge top band stitching shall be performed after the lining and the top of the eyelet stays are inserted. The first row of outside stitching shall be 1/16 to 1/8 inch from the top band edge. These two rows of stitching may be performed with a double needle machine.".

Sentences 19 and 20 - Delete all references to "speed lacer" and substitute "eyelet".

Sentence 21 - After "crimped" add ", if necessary,".

PAGE 14

*3.7.4.3, line 4 - Delete "succunic" and substitute "succinic".

Line 5 - Delete "speed lacer" and substitute "eyelet".

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*3.7.5 - Delete in its entirety and substitute "3.7.5 Eyeletting. Seven eyelets (eight for sizes 9 and larger) shall be fastened to the boot in accordance with contractor supplied marker patterns or any other suitable positioning method. Each quarter shall contain seven eyelets (eight for sizes 9 and larger). The outer edge of the eyelets shall be not less than 3/16 inch or more than 5/16 inch from edge of quarter. The top eyelet shall be positioned just far enough below the bottom edge of the top band to miss going through the band piece. The bottom eyelet shall be positioned so that its outer edge is approximately 1/4 inch from edge of quarter at the blucher. The remaining eyelets shall be evenly spaced and properly aligned.".

PAGE 15

*3.7.7, line 2 - Delete "lacing holes." and substitute "sets of eyelets, including the bottom set of eyelets.".

*3.7.8, line 5 - Delete "lacer" and substitute "eyelet". Delete "figure 1" and substitute "Figure 1A".

*3.7.10, sentence 3 - Add the following after sentence 3 "When automatic backpart lasting equipment is used, the tack at the center of the heel seat may be omitted.".

PAGE 16

*3.7.16, line 2 - Delete "a suitable adhesive." and substitute "an adhesive as specified in 3.3.14.2.".

PAGE 17

3.7.18, line 5 - Delete "4-1/2" and substitute "5-1/2".

*3.7.21, line 1 - After "nails" add "however, heels size 5 and below shall be attached with 7 nails".

PAGE 18

*3.7.27, sentences 2, 3 and 4 - Delete in their entirety and substitute "A lace shall be inserted through the top eyelet on the inside quarter of each boot and both laces tied together firmly.".

*3.8.2 - Delete in its entirety and substitute: "3.8.2 ANSI labeling. The boots shall be labeled to indicate that they conform to class 75 safety-toe requirements outlined in American National Standards Institute Standard ANSI Z41.1. Each boot shall be stamped with a permanent marking ink of contrasting color or shall have an indelible cloth label stitched to the inside of the boot on one of the quarter linings or on upper part of tongue on the flesh side. The numbers and letters shall be a minimum of 3/16 inch in height and the label inscription shall read as follows:

ANSI Z41 PT 83
M I/75 C/75

PAGE 19

*3.8.3 - Delete in its entirety.

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PAGE 21

*Table VI, Upper leather - After paranitrophenol content delete "(when applicable)".

After lining leather and heel pads - Add the following in appropriate columns: "Paranitrophenol content, 3.3.1.2, FED-STD-311, 6711 1/".

PAGE 24

*Table VI - Delete "Speed lacers" and "Clinch clamp" and subsequent test procedures.

PAGE 25

*Table VI - Delete "Adhesive general" and subsequent test procedures and add the following new test procedure in the appropriate columns:

"General Adhesive (1 gallon)	Welt to Outsole Initial	3.3.14.2	D-816, B& 3.3.14.21/	X	- 6	psi	6 cemented units 4/
	After Immersion	3.3.14.2	D-816, B& 3.3.14.21/	X	- 6	psi	6 cemented units 4/"

*Table VI - Add the following footnote "4/ The six assemblies shall consist of welting strips joined to outsoles. Each specimen shall have a minimum length of six incnes. The width of the test specimen shall be sufficient to assure complete adhesion to the welting strip. The flat side of the welt shall be bonded to the outsole. If failure occurs other than along the adhesive line, the value shall not be scored."

PAGE 26

*4.3.2.1, line 7 - After "cut" add "showing through on grain surface". Add the following "Uppers with fat wrinkles, pits, stretchy leather, insect or surface damage".

PAGE 27

*4.3.4.1.1, Table of Defects - After construction and workmanship defect e add "when necessary".

Under "Quality of leather" - After defect a, add "(Upper leather only)".

- After defect d, add "on grain surface".

- After defect f, add the following:

"g. Excessive variation in color of lining leather within a boot - minor
h. Excessive variation in color of lining leather with a pair - minor
i. Fat wrinkles, pits, stretchy leather, insect or surface damage in uppers - major
j. Flanky, fleshy, loose, boney, pipey, cracked, pronounced veins, rough or coarse grain on upper leather - major
k. Flesh cuts, flanky or pipey leather affecting serviceability of lining - major"

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PAGE 32

Table of Defects, under outsole stitching and bottom attaching (Goodyear):

Line 4 - Delete "3-1/2" and substitute "4-1/2".

Line 6 - Delete "4-1/2" and substitute "5-1/2".

Line 7 - Delete "3-1/2" and substitute "4-1/2".

Under Edgemaking defect a - delete "major" and substitute "*".

PAGE 33

*Table of Defects, under "Lacers and eyelets" - delete title and substitute "Eyelets".

Defect a - delete "lacers or".

Defect b - delete "Lacers" and substitute "Eyelets".

Defect e - delete in its entirety and substitute
"e. Clinched

1. Tightly, damaging or cutting adjacent material - minor
2. Loosely, permitting component to rotate freely, but not to the degree that any component can be expected to become detached during use - minor
3. Loosely, to the degree that components can be expected to become detached during use - major".

Defect f - delete "Lacer" and substitute "Eyelet".

Defect g - delete in its entirety.

Defect h - delete "lacer or".

PAGE 34

*After defect table, footnote 2, add "The contractor shall perform 100% examination for this defect.".

4.3.5.2, after sentence 4 add "Sample units for the impact test shall be selected by random numbers and shall consist of footwear sizes chosen from the entire range of the tariff.".

PAGE 35

*4.3.6, sentence 6 - Delete the period and add "in accordance with MIL-STD-105.".

Add the following new paragraph:

"4.3.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

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PAGE 35 (Cont'd)

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.".

PAGE 37

*4.4.4, line 21 - Delete "1976" and substitute "1983".

PAGE 38

5.2.1, sentence 2 - Delete "weather-resistant, variety DW, grade V15C" and substitute "domestic."

Sentence 3 - Delete in its entirety and substitute "Two different size containers, one for the smaller sized boots and one for the larger sized boots, may be used. The approximate interior dimensions are given in inches for the two containers:

Boot sizes	Length	Width	Depth
Up thru 9-1/2	25	13	10-1/4
10 and above	26	14	11-1/2

The contactor, based on his schedule of sizes, shall determine the actual size of containers to be used.".

PAGES 38 - 39

5.2.2, Sentence 2 - Delete "variety DW, grade 275".

Sentence 3 - Delete in its entirety and substitute "Two different size containers, one for the smaller sized boots and one for the larger sized boots, may be used. The approximate interior dimensions are given in inches for the two containers:

Boot sizes	Length	Width	Depth
Up thru 9-1/2	25	13	10-1/4
10 and above	26	14	11-1/2

The contactor, based on his schedule of sizes, shall determine the actual size of containers to be used.".

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PAGE 39

5.3.1, line 3 - Delete "Sizes and Widths" and substitute "NSN'S."

line 4 - After "legibly" insert "stamped or ".

line 5 - Delete "of pairs, sizes and widths" and substitute "and national stock numbers".

Add the following new paragraph after 5.3.1:

*5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1a of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 from wood groups specified in MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer."

*6.2 Add the following: "f. When palletization is required (see 5.4).".

PAGE 40

*After 6.6 add the following new paragraphs:

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.8 Subject term (key word) listing.

Boot, blucher style
Leather (water resistant)
Soles, rubber fuel resistant
Toe, safety steel."

PAGES 41 and 42

*Figures 1 and 2 - Delete the two figures and insert new figure 1A.

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT
Navy - AS

Project No. 8430-0363

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Figure 1A

BOOTS, SAFETY, AIRCREW

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